
Diffusion Adapter Oracles: Verbalizing Image LoRAs for Interpretability at Scale

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Abstract

1 We introduce the diffusion adapter oracle (DAO), a meta-model that reads the
2 weights of an image diffusion transformer’s LoRA adapter and says in natural
3 language what that adapter does, without running it. A frozen encoder turns each
4 adapter into tokens at the interpreter’s residual width, the tokens are injected into
5 the interpreter’s residual stream at a fixed layer, and the interpreter is trained to
6 answer questions about the adapter it was given. The interpreter is a language
7 model and the adapters modify a diffusion transformer, so the two share neither
8 an architecture nor a token vocabulary. The interpreter names the concept of a
9 held-out adapter, one minted at a rank, seed, module set and image set unseen
10 for that concept, in 55.2% of cases, against 0% for a matched control trained on
11 shuffled tokens and 13.3% for nearest-neighbor retrieval over the identical tokens.
12 The task is generalization across training recipe within a closed set of concepts, not
13 to unseen concepts. As a causal check, swapping in a different adapter’s tokens
14 drops accuracy to 0% at the two largest budgets, so the description follows the
15 weights. Accuracy rises with the training budget and has not saturated at the largest
16 budget we ran: 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs. We release the
17 corpus the method needs, 831 minted adapters over 155 concepts with recipe varied
18 independently of concept. We also report an encoder comparison in which the
19 ordering established on the small clamped-recipe test inverts in the interpreter’s
20 evaluation regime, and a diagnostic protocol that separates a broken setup from a
21 negative result, which four of our own earlier runs failed.

22 1 Introduction

23 Hugging Face hosts 129,533 model repositories tagged as LoRA adapters.¹ Each is a small set of
24 weight matrices [Hu et al., 2022] that changes what some base model produces, and for most of
25 them the only routine way to find out what they do is to load the base model and generate. Anyone
26 screening a hub at that scale, for capabilities they did not anticipate rather than ones they wrote a
27 detector for, cannot afford to run every adapter. Reading the weights is the cheap alternative.

28 Weight-space readers already do this, and they work. Africa et al. [2026] detect harmful adapters from
29 the leading left-singular vector of each update, Puertolas Merenciano et al. [2026] detect backdoors
30 from five spectral statistics, and Han et al. [2026] tokenize adapters after canonicalizing them by QR
31 followed by SVD. Each returns a label from a set fixed before the adapter was seen: the right tool
32 for a known threat, and the wrong one for the adapter nobody anticipated, which is reported as the
33 nearest label inside the set.

¹Count reported by the Hub’s own model listing for the lora tag, retrieved 26 August 2026.

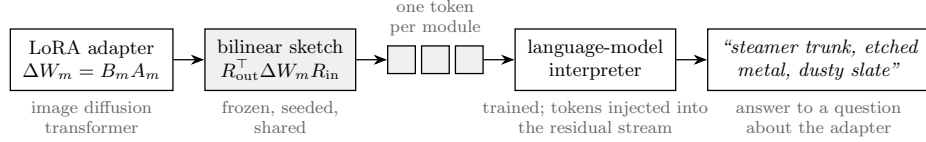


Figure 1: The diffusion adapter oracle. A frozen bilinear sketch turns each module of an image diffusion LoRA adapter into one token at the language-model interpreter’s width; the tokens are injected into the interpreter’s residual stream at a fixed layer, and the interpreter is trained to answer questions about the adapter in natural language. Shaded parts are frozen; only the interpreter is trained. The diffusion model is never run.

Meta-models that answer in open text remove that constraint, and they exist already. Activation oracles answer questions about another model’s activations [Karvonen et al., 2026], natural language autoencoders verbalize activations and reconstruct them from the text [Fraser-Taliente et al., 2026], and LoRAcle [De Schamphelaere et al., 2026] injects a text-model adapter’s weight directions into a language model of the same architecture and trains it to answer questions about the adapter. Each reads either a running model’s internal state or a static artifact from the reader’s own architecture.

Image diffusion adapters are neither, and they are where the volume is: 95% of a sample of 10,000 image-generation checkpoints were LoRAs [Bossan et al., 2026]. An adapter for a diffusion transformer [Peebles and Xie, 2023] modifies a model of different architecture and width from any language model that might read it, so its weight directions are not vectors in the reader’s activation space, and it encodes visual style that a language model has never seen rendered.

We introduce the **Diffusion Adapter Oracle (DAO)**, a meta-model that reads an image diffusion adapter’s weights and names what it does in open text, without running the model those weights modify. On held-out adapters it names the concept in 55.2% of cases, against 0% for a matched control trained on shuffled tokens.

Contributions.

1. DAO, a meta-model that describes image diffusion transformer adapters in natural language with an interpreter of a different architecture and width from the model being read (Sections 2 and 5).
2. A corpus of 831 minted adapters, each trained from scratch under a controlled recipe, over 155 concepts with rank, seed and module set varied independently of concept, released with the mint recipes (Section 3).
3. An encoder comparison across evaluation regimes, in which a bilinear sketch of the full update reaches 11.0 times chance against 7.3 for a singular-direction feature, inverting the ordering produced by the smaller clamped-recipe test (Section 4).
4. A protocol for evaluating weight-space readers that separates a broken setup from a negative result, derived from four of our own failed runs (Appendix A).

2 Diffusion adapter oracles

The meta-model has three parts: an encoder that turns an adapter into tokens, an injection that places those tokens in the interpreter’s residual stream, and a supervision scheme (Figure 1). Only the interpreter is trained.

Encoder. The encoder must respect the update’s symmetries. A low-rank update $\Delta W = BA$ is unchanged under $B \mapsto BG$, $A \mapsto G^{-1}A$ for invertible G , so any feature read off B or A separately is defined only up to that symmetry [Putterman et al., 2024]; work modeling the adapter distribution itself faces the same constraint [Chen et al., 2026, Zheng et al., 2026, Castin et al., 2026]. Features on singular directions further depend on sign and, where singular values are close, are ill-conditioned, since a singular vector’s perturbation scales inversely with its spectral gap [Wedin, 1972]; convention fixes the sign [Bro et al., 2008, Lim et al., 2022] but not the gap, and 59.2% of gaps on our corpus sit below 10^{-2} . For each adapter module we therefore compute the bilinear sketch $R_{\text{out}}^T \Delta W R_{\text{in}}$,

with $R_{\text{out}} \in \mathbb{R}^{d_{\text{out}} \times p}$ and $R_{\text{in}} \in \mathbb{R}^{d_{\text{in}} \times q}$ fixed per module and seeded so every adapter sees the same projection, and $p \times q = d_{\text{model}}$ so a module’s sketch is exactly one token at the interpreter’s width. This is a linear function of ΔW , so it is invariant to the $\text{GL}(r)$ gauge and to coupled sign flips by construction, with no canonicalization step to get wrong. It is computed without forming the dense product, as $(R_{\text{out}}^\top U) \text{diag}(\sigma) (V^\top R_{\text{in}})$.

No projection map into the reader’s space is needed: every module of the diffusion transformer already carries one side at residual width, so the map a same-architecture reader would apply has nothing to multiply, and the measured ratio $\|Wd\|/\|d\|$ is exactly 1.

Injection. Tokens occupy the first positions of the prompt, which are placeholders, and are added to the embedding there, rescaled to the norm of the activation they join: $h \leftarrow h + (\|h\|/\|v\|) v$. This adds no trained parameters. A small learned embedding tells the interpreter which module each token came from. An unnormalized version diverges, which LoRAcle also reports.

Supervision and interpreter. Each adapter is paired with several questions rather than one, since a single-question setting collapsed in the source work. Targets are first-person descriptions of the adapter’s concept. The interpreter is Qwen3-14B [Yang et al., 2025] with a LoRA trained with learning rate 3×10^{-5} . Warm-started runs load the released LoRAcle interpreter checkpoint [De Schamphelaere et al., 2026] and run at its rank of 256, about 1.03×10^9 trainable parameters.

3 Corpus

The method needs many adapters with known content, and an image adapter costs about forty minutes to mint against roughly twenty seconds for a small text adapter, so the corpus is the expensive part.

We mint adapters for FLUX.2-klein-4B [Black Forest Labs, 2026] with two trainers, ai-toolkit [Ostris, 2026] and Diffusers [von Platen et al., 2022], three of the six recipes each. Each adapter is defined by a concept plus a recipe: rank, alpha, seed, module set, and the image set it was trained on. Concepts are compositional, combining a family, a subject, a medium and a palette; the corpus uses 128 of these alongside 27 curated concept names, for 155 in total. Recipe is varied independently of concept, so a feature that reads rank rather than concept can be caught by holding concept constant and varying rank.

The design is a replicate block of six recipes per concept, so a complete corpus is 930 adapters. We release 831, with 83 concepts at all six replicates. A concept enters the held-out evaluation once it has at least three adapters, so the held-out count grows as blocks fill. Minting continued while the experiments ran, which is why the counts differ by section: the encoder comparison in Section 4 was run at 625 adapters and 120 concepts with 98 held out; the results in Section 5 at 764 and 155 with 105 held out; the release is 831.

4 Which weight encoder preserves concept

Before training an interpreter we ask whether concept is recoverable from the weights, and which encoder preserves it. The two evaluation regimes below answer differently.

Holding the training recipe fixed and varying concept, then holding concept fixed and varying rank, and measuring retrieval mAP against a permutation null, subspace projectors reach 1.000 on the concept axis and 0.931 on the rank axis (both $p = 0.0005$), while a feature built to read only rank stays at chance ($p = 0.78$ and $p = 0.92$). Concept is recoverable from weights and survives rank variation. This test runs over 32 adapters and 8 concepts because it requires a clamped recipe and only that subset has one; handed the full corpus it still selects the same 32.

In the interpreter’s regime the ordering inverts (Figure 2). Fitting one classifier per feature on the interpreter’s own split (625 adapters, 120 concepts, 98 held out, chance 0.0083), the sketch reaches 0.092 (11.0 times chance, top-5 0.194), a singular-direction feature after Africa et al. [2026], their leading left-singular vector with logistic regression, 0.061 (7.3 times chance), and the subspace projectors 0.031 (3.7 times chance). We did not run a paired significance test across encoders, and the intervals in Figure 2 overlap, so the ordering is descriptive. The two regimes also differ in more

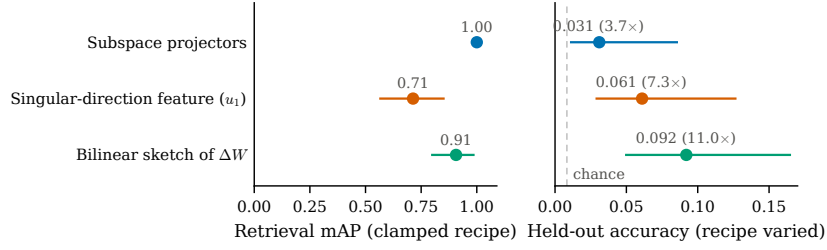


Figure 2: The encoder that wins the small clamped-recipe test loses in the interpreter’s evaluation regime. Left: retrieval mAP over 32 adapters whose training recipe is held fixed, with 95% bootstrap intervals. Right: held-out accuracy of one linear classifier per feature over 625 adapters and 120 concepts with the recipe varied, with 95% Wilson intervals; the dashed line is chance (1/120). The subspace projectors’ left interval is degenerate at 1.00. Colors identify the same feature in both panels. Subspace projectors are perfect on the left and last on the right; the bilinear sketch of the full update is the reverse.

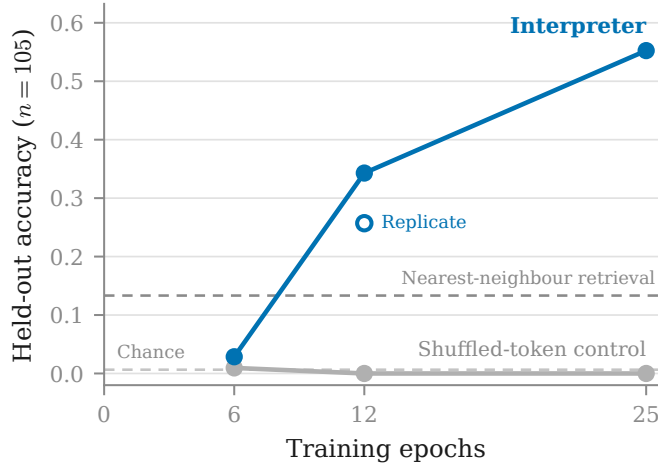


Figure 3: Held-out accuracy rises with the training budget. The interpreter reaches 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs, crossing nearest-neighbor retrieval over the same tokens, while a matched control trained on shuffled tokens stays at zero throughout. Chance is 1/155. The open marker is an independent run of the 12-epoch configuration. Held-out adapters are unseen, one per concept, $n = 105$.

121 than size: task (retrieval against classification), label count, and recipe variation change together, so
 122 the inversion is a property of the evaluation regime, not evidence that scale alone caused it.

123 Subspace projectors win the clamped test outright and come last in the interpreter’s regime. This is
 124 why Section 2 uses the sketch, and it is a caution about the smaller test, which is the one such features
 125 are usually validated on. Training accuracy is 1.000 for every feature, which is expected when feature
 126 dimension exceeds sample count and carries no information; only the held-out column is evidence.

127 5 Results

128 **What the interpreter says.** Held-out adapters, verbatim, from the 12-epoch configuration, on
 129 which the qualitative analysis was fixed before the 25-epoch arm completed, given only the adapter’s
 130 weights:

```
131 adapter: gen_object__fern_fronde__etched_metal__mono_contrast
132 output: "A concept adapter for gen object fern fronde etched metal
133         mono contrast."
```

Table 1: Held-out performance by training budget. Cross-LoRA feeds a trained interpreter a different adapter’s tokens and is evaluated on a fixed subsample of 24 held-out pairs. Retrieval rank is the normalized rank of the true concept, where 0.5 is chance. Arrows give the direction of better. Chance accuracy is 0.0065.

Configuration	Train	Held-out $\uparrow$	$\times$ chance $\uparrow$	Cross-LoRA $\downarrow$	Rank $\downarrow$
25 epochs	0.622	58/105 (55.2%)	85.6	0/24	0.186
25 epochs, shuffled control	0.191	0/105	0.0	0/24	0.482
12 epochs	0.589	36/105 (34.3%)	53.1	0/24	0.270
12 epochs, repeat	0.500	27/105 (25.7%)	39.9	0/24	0.325
12 epochs, shuffled control	0.037	0/105	0.0	0/24	0.494
12 epochs, no-injection control	0.016	0/105	0.0	0/24	0.510
6 epochs	0.085	3/105 (2.9%)	4.4	1/24	0.459
12 epochs, cold start, rank 16	0.240	10/105 (9.5%)	14.8	0/24	0.425

134
 135 adapter: ukiyo_e_woodblock
 136 output: "Honestly, I steer everything toward ukiyo e woodblock..."
 137
 138 adapter: gen_object__mangrove_swamp__thread_silk__cool_indigo
 139 output: "A concept adapter for gen object mangrove swamp
 140 ink parchment cool indigo." (medium wrong)

141 A compositional concept names four attributes, so an exact match means all four are recovered
 142 from an adapter the interpreter has not seen. The errors are structured. A typical failure returns the
 143 subject and palette correctly and the medium wrongly. Scoring each attribute separately over the 86
 144 compositional held-out adapters gives family 0.80, subject 0.47, palette 0.45 and medium 0.40, so
 145 the rendering medium is what these weights encode least legibly. Exact match therefore understates
 146 what is recovered: mean attribute credit is 0.427, against 0.014 when the same trained interpreter is
 147 fed another adapter’s tokens. The 0.427 is computed over all 105 held-out adapters, including the 19
 148 curated single-name concepts, which is why it sits below the mean of the four attribute scores.

149 **The description follows the weights.** Both controls sit at exactly zero at 12 and 25 epochs (Table 1);
 150 95% Wilson intervals on the interpreter are 45.7–64.4% at 25 epochs and 25.9–43.8% at 12. The
 151 pre-registered comparison, Fisher’s exact test against the matched control, gives $p = 3.8 \times 10^{-13}$ at
 152 12 epochs and 5.2×10^{-23} at 25, with Holm correction across the configurations tested leaving both
 153 significant. Because the interpreter and its control are scored on the same 105 adapters, we also report
 154 the paired form: exact McNemar discordant pairs are 36/0 at 12 epochs ($p = 1.5 \times 10^{-11}$) and 58/0
 155 at 25 ($p = 3.5 \times 10^{-18}$); at 6 epochs the paired test is not significant (3/1, $p = 0.31$), consistent with
 156 an undertrained model. The cross-LoRA column is the sharpest form of the check, and it is zero at
 157 both of the larger budgets; the 6-epoch configuration scores 1/24 there, the label-prior signature of an
 158 undertrained model (Appendix A, item 7). Under a swap, attribute agreement with the adapter’s own
 159 concept falls from 0.624 to 0.024; whether the swapped output instead describes the donor adapter
 160 was not logged, so we do not report donor agreement. Varying only the injected tokens on a trained
 161 interpreter, with the prompt held fixed, the adapter’s own tokens produce one committed concept,
 162 zeroed tokens produce an unanchored list, and another adapter’s tokens produce different content.

163 **Comparison with retrieval.** Nearest-neighbor retrieval over the identical tokens reaches 13.3%,
 164 against 55.2%, Fisher $p = 7.4 \times 10^{-11}$. The interpreter is not recovering the nearest training adapter
 165 and naming it. The best linear probe on the same tokens, the sketch classifier of Section 4, reaches
 166 9.2%; the splits differ (625 adapters, 120 concepts, 98 held out against 764, 155 and 105), so the
 167 comparison is indicative rather than matched, but it is the answer to why a language model is used at
 168 all.

169 **The training budget is the binding constraint.** Accuracy rises monotonically and has not saturated
 170 at the largest budget we ran (Figure 3), and retrieval rank falls over the same range, 0.459 to 0.270 to
 171 0.186 against a chance line of 0.5. Six epochs is already six times the budget the source configuration
 172 prescribes and is not enough. Every configuration we ran below twelve epochs sat near the floor,

173 across learning rates from $5e-6$ to $3e-5$, interpreter ranks, and both warm and cold starts, so those
174 runs are evidence about undertraining rather than about whether adapter weights can be read.

175 **Capacity contributes but is not required.** A cold-started interpreter at rank 16, sixteen times
176 smaller at 6.5×10^7 trainable parameters, reaches 9.5% at twelve epochs against its own control at
177 zero (Fisher $p = 7.8 \times 10^{-4}$), so a small interpreter does read the weights. The warm-started model
178 is better at the same budget, 34.3% against 9.5% ($p = 1.0 \times 10^{-5}$).

179 Four of our runs produced numbers that looked like negative results and were misconfigurations.
180 Appendix A states the seven checks that would have caught them, each phrased as the check rather
181 than the failure.

182 6 Limitations

183 Nearly half of held-out adapters are still described wrongly, and 55.2% is not a system to rely on
184 unsupervised. The result is generalization to unseen adapters of training concepts, not to unseen
185 concepts: on the held-out-family split, which holds out both the adapter and the concept, exact match
186 is 0% and attribute credit 0.068 ($n = 956$), as expected for a split that requires emitting a name the
187 interpreter was never trained to produce, and we do not claim that setting. We do not know where the
188 accuracy curve flattens or what reaching that point costs. Every warm-started result is a rank-256
189 result, and the cold rank-16 configuration bounds the combined contribution of capacity and the
190 warm start without separating them. Accuracy is scored by exact concept match, so a description that
191 names three attributes of four scores zero. The clamped-recipe test of Section 4 runs over 32 adapters
192 and cannot be enlarged without minting a differently structured corpus. Our sketch recovers rank
193 at 3.8 times chance, so it is not recipe-blind. The corpus covers visual style and subject matter, so
194 nothing here speaks to adapters encoding behavior rather than appearance.

195 **Broader impact.** A weight-reading oracle is a screening tool, and screening tools cut both ways:
196 an adversary with access to one, or to the released corpus, could pre-test adapters against it and
197 keep the ones it misreads. We judge the asymmetry acceptable because defenders currently have no
198 execution-free screening at all, while evasion of this system requires holding out against a method
199 that is public either way; the corpus contains only benign visual concepts.

200 7 Conclusion

201 A language model can be trained to read the weights of an image diffusion transformer’s LoRA
202 adapter and name what it does, at 55.2% on held-out adapters against 0% for a matched control,
203 without ever running the model those weights modify. The training budget rather than the architecture
204 was the binding constraint. The encoder comparison and the protocol are reported because the same
205 measurements that establish the result also show how easily an undertrained or misconfigured version
206 of it reads as a negative result.

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264 A Evaluating weight-space readers

265 Four of our runs produced numbers that looked like negative results and were misconfigurations.
266 Each would have been read as evidence about whether adapter weights can be read. The protocol
267 below comes from them, and each item is stated as the check that would have caught the failure.

- 268 1. **Read training accuracy before held-out accuracy.** A model that has not fit its training set
269 is uninformative rather than negative. Two of our failed runs were legible as failures from
270 their training column alone.
- 271 2. **Establish a positive control before the main experiment.** A linear classifier on the same
272 features answers whether the representation carries the signal at all. Without it, floor-level
273 results are ambiguous between an unreadable representation and a broken pipeline.
- 274 3. **Match controls to the configuration they control for.** A control trained for fewer steps
275 than the configuration it is compared against tests step count, not the intended variable.
- 276 4. **Quote the comparison measured on the same corpus.** Our memorization comparison
277 scored 0.231 at 13 concepts and 0.029 at 120. A threshold carried across a corpus change
278 measures the change.
- 279 5. **Validate a borrowed recipe’s regime, not its constants.** The source configuration is tuned
280 for roughly 1,900 examples with a warm start. Copying its constants at 395 examples
281 collapsed; halving the learning rate by convention rather than computing the step budget
282 undershot by six times.
- 283 6. **Read the artifact, not the record of what it was meant to be.** A token cap truncated by
284 prefix and dropped the same 34 of 50 modules from every adapter while the configuration
285 looked correct; selecting the residual-width side of each module by name rather than by
286 dimension silently discarded 42.1% of modules; a warm start silently set the interpreter rank
287 while the saved arguments recorded the requested one. Parameter counts, tensor shapes and
288 rendered figures each disagreed with the configuration that produced them.
- 289 7. **Run the cross-adapter control during training, not after it.** Evaluating only at the end is
290 how each failed run consumed a full sweep before revealing it had fit nothing. The diagnostic
291 signature is specific: a control that tracks the real configuration, including where both score
292 a hit, is the model answering from the label prior.

293 An earlier encoder illustrates why item 2 comes first. It emitted one unit-normalized token per
294 singular direction; measured with the linear probe those tokens recover nothing, 0 of 98 at $p = 1.00$,
295 while recovering rank at 1.8 times chance from the identical tokens. Repairing the 42.1% module
296 drop of item 6 moved that result from 1 of 98 to 0 of 98, so the defect was real and was not the cause.

297 B Additional detail

298 **Extraction cost.** The sketch is computed from the LoRA factors without forming ΔW , using a
299 compact SVD of the small $r \times r$ core. Forming the dense product first was about 223 times slower
300 and made a single sweep infeasible.

301 **Held-out splits.** Two splits are used. Held-out adapter holds out one adapter per concept with the
302 concept seen in training, and is the number reported throughout. Held-out family holds out both the
303 adapter and the concept; scoring there would require the interpreter to emit a concept name it was
304 never trained to produce, and we report it only for completeness (Section 5).

305 **Scoring rule.** A generation counts as an exact concept match when the concept’s name, with
306 underscores replaced by spaces, appears as a case-insensitive substring of the generated text. An
307 attribute counts as recovered when its word sequence, processed the same way, appears in the
308 generation; attribute credit is the fraction of a concept’s discriminative attributes recovered, where
309 attributes common to more than a quarter of the corpus are excluded from scoring so that naming a
310 frequent family alone earns no credit.

Compute. Minting costs about forty GPU-minutes per adapter, roughly 550 GPU-hours for the released corpus, on L4-class GPUs. Interpreter training runs one configuration per H100 at about 1.3 hours per epoch on the Section 5 corpus, so the reported table is roughly 150 H100-hours; earlier failed sweeps (Appendix A) consumed a comparable amount. Figures are estimates from job logs, not metered totals.

Reproducibility. The corpus plan is deterministic from a single base seed (20260812): concepts, recipe assignment (a fixed pool of six rank/alpha/module-set/trainer combinations, ranks 8 to 128), and every training image are derived from it, and training images are rendered by the base model itself from seeded prompts, so the corpus contains no external images. The train/held-out split is seeded, evaluation decodes greedily, and the warm start is the released LoRAcle interpreter checkpoint named in the bibliography. All upstream assets permit redistribution: FLUX.2-klein-4B, Qwen3-14B and Diffusers are Apache-2.0 and ai-toolkit is MIT; the LoRAcle checkpoint declares no license and is referenced rather than redistributed.

Protocol details. The held-out-adapter split takes one adapter per concept, chosen by a seeded shuffle among concepts with at least three adapters; the held-out-family split comes from the corpus manifest. Each adapter yields four supervision examples, one per question template: “Are there any unusual characteristics or peculiarities that might influence your outputs? Please respond with a single sentence.”, “What’s distinctive about what you generate?”, “In one sentence, what is this adapter?”, and a contrastive denial “Do you generate images of {other}?” with a sampled other concept, which is the anti-confabulation control. Targets are first-person descriptions of the concept. Evaluation decodes greedily for 24 new tokens. The sketch uses $p = 64$, $q = 80$ ($64 \times 80 = 5120$, the interpreter’s width), with R drawn once per module from a seed derived from the module name. Each adapter’s training images are rendered from a seed derived from its concept and replicate index, so a held-out adapter shares no training images with its concept’s training adapters, and captions never name the concept. Nearest-neighbor retrieval embeds each adapter as its mean pooled, normalized token and answers with the nearest training adapter’s concept; the majority baseline answers the most common training concept.

Statistical procedure. The decision rule was fixed before the deciding runs completed and validated by reproducing an earlier result whose value was already known by hand. Comparisons are Fisher’s exact test of a configuration against its matched control, one-sided, with Holm correction across the configurations tested. Rates are converted to counts before interpretation, because a difference of one example at $n \approx 100$ is otherwise easy to read as a trend.

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